UNIVERSITY PHYSICS REFERENCE SERIES • VOLUME 6

Fluid Mechanics: Statics, Dynamics & Viscous Transport

Hydrostatic Pressure Fields, Pascal's Principle & Hydraulic Machinery, Archimedes' Buoyancy, Continuity Equation, Bernoulli's Principle & Comprehensive Applications (Venturi Meter, Torricelli Efflux, Pitot Tubes, Aerodynamic Lift), Hagen-Poiseuille Viscous Flow, Reynolds Turbulence Criteria, and Hand Calculation Fast-Tracks.

Synthesized From:

- University Lecture Slides — *Chapter 5: Fluids*
- Textbook — *Fundamentals of Physics: Mechanics, Relativity, and Thermodynamics* (Chapter 20)
- Reference Formula Sheet — *Formula Phy1* (Equations Embedded Directly)

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Features Included: 4 Scalable Vector SVG Illustrations • 5 Calculation Fast-Tracks • 5 Step-by-Step Worked Problems

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Part I: Fluid Statics & Hydrostatic Pressure Fields

Chapter 1: Foundations of Fluids, Density & Continuum Hypothesis

In classical continuum mechanics, matter is categorized into solids and fluids based on its response to applied mechanical stress. While an elastic solid deforms by a finite, static strain angle when subjected to an external shear stress (returning to its resting state upon removal of the stress), a **fluid** (encompassing both liquids and gases) is defined as a substance that *deforms continuously and indefinitely under the application of any shear (tangential) stress*, no matter how small that stress may be. Under static conditions, a fluid cannot support shear stress; consequently, static fluids exert only normal forces on any confining boundary.

1.1 The Continuum Hypothesis

Although matter consists microscopically of discrete atoms and molecules separated by empty space, engineering and macroscopic physics treat fluids as continuous fields. The validity of this **continuum hypothesis** is governed by the *Knudsen number*, defined as:

$$\mathbf{Kn} = \frac{\lambda}{L}$$

where λ represents the molecular mean free path (the average distance between successive molecular collisions, ≈ 68 nm in atmospheric air) and L denotes the characteristic macroscopic length scale of the physical system (e.g., pipe diameter or airfoil chord). Whenever $\mathbf{Kn \ll 0.01}$, macroscopic averages of molecular momentum and mass density become smooth, continuous functions of spatial coordinates (x, y, z) and time t: density field $\rho(\mathbf{r}, t)$, velocity field $\mathbf{v}(\mathbf{r}, t)$, and pressure field $P(\mathbf{r}, t)$.

1.2 Definition of Mass Density

The **mass density** ρ of a fluid element is defined as the ratio of its infinitesimal mass Δm to its infinitesimal volume ΔV as ΔV shrinks to a microscopic continuum limit δV^*:

$$\rho = \frac{dm}{dV} = \frac{m}{V} \quad \textbf{[for uniform substances]}$$

In the International System of Units (SI), density is measured in **kilograms per cubic meter ($\mathbf{kg/m^3}$)**. In laboratory settings and chemistry, density is frequently quoted in grams per cubic centimeter (g/cm^3). The exact conversion factor is:

$$1\ \mathbf{g/cm^3} = \frac{10^{-3}\ \mathbf{kg}}{(10^{-2}\ \mathbf{m})^3} = 1000\ \mathbf{kg/m^3}$$

Substance	Phase / State	Density ρ (kg/m^3)	Specific Gravity (SG)
Pure Fresh Water (4 °C)	Liquid (Standard)	1,000	1.000
Sea Water (Saline, 3.5%)	Liquid	1,025	1.025
Whole Human Blood (37 °C)	Liquid	1,060	1.060
Mercury (Hg, 20 °C)	Liquid metal	13,600	13.60
Ethyl Alcohol (Ethanol)	Liquid	789	0.789
Standard Atmospheric Air (20 °C, 1 atm)	Gas mixture	1.204	0.00120

Substance	Phase / State	Density ρ (kg/m^3)	Specific Gravity (SG)
Pure Ice (0 °C)	Solid	917	0.917

The **Specific Gravity (SG)** (or relative density) is a dimensionless ratio comparing the density of a substance to the standard density of pure water at its maximum density temperature of 3.98 °C (ρ_W = 1000 kg/m^3):

$$SG = \frac{\rho}{\rho_{water,\ 4°C}} = \frac{\rho\ [kg/m^3]}{1000\ kg/m^3}$$

Chapter 2: Hydrostatic Equilibrium & Pressure Variation with Depth

When a fluid remains at rest relative to an inertial reference frame, all shear stresses vanish. The only internal mechanical interaction between adjacent fluid parcels is a normal compressive force.

2.1 Definition and Units of Pressure

Consider an infinitesimal planar surface of area dA immersed within a fluid or located on the container boundary. If the fluid exerts a purely normal compressive force $dF_\perp$ across this area, the **hydrostatic pressure P** is defined as:

$$P = \frac{dF_\perp}{dA}$$

Pressure is an *intensive scalar quantity*. At any given point within a static fluid, pressure acts with equal magnitude in every spatial direction (proven via static equilibrium of an infinitesimal tetrahedral fluid element). The SI derived unit of pressure is the **Pascal (Pa)**, defined as one Newton per square meter:

$$1\ Pa = 1\ N/m^2 = 1\ kg\ /\ (m \cdot s^2)$$

Because one Pascal represents a very small pressure (equivalent to the weight of a sheet of paper resting on a table), several non-SI units are routinely used across physics and engineering:

- **Standard Atmosphere (atm):** Defined as the mean sea-level atmospheric pressure: **1 atm = 1.01325 × 10^5 Pa = 101.325 kPa ≈ 1.013 × 10^5 N/m^2**.
- **Bar (bar): 1 bar = 10^5 Pa = 100 kPa ≈ 0.9869 atm**.
- **Millimeters of Mercury (mmHg / Torr):** The hydrostatic pressure exerted by a 1 mm vertical column of liquid mercury at 0 °C: **760 mmHg = 1 atm ⇒ 1 mmHg = 1 Torr = 133.322 Pa**.
- **Pound per Square Inch (psi): 1 atm ≈ 14.696 psi ⇒ 1 psi ≈ 6,894.76 Pa**.

2.2 Derivation of the Fundamental Hydrostatic Equation

To determine how pressure varies through a fluid column in a uniform gravitational field $\mathbf{g} = -g\,\hat{\mathbf{j}}$, isolate an infinitesimal cylindrical fluid element of horizontal cross-sectional area A and vertical thickness dy located between elevations y and y + dy.

The three vertical forces acting on this fluid element in static equilibrium are:

1. Upward force on the lower face at elevation y: $\mathbf{F_{bottom} = P(y) \cdot A}$
2. Downward force on the upper face at elevation y + dy: $\mathbf{F_{top} = P(y + dy) \cdot A = [P(y) + (dP/dy)\,dy] \cdot A}$
3. Downward gravitational weight force: **dW = dm · g = ρ · dV · g = ρ A dy · g**

Applying Newton's first law of static equilibrium ($\sum F_y = 0$):

$$P(y)\,A - [P(y) + \frac{dP}{dy}\,dy]\,A - \rho\,g\,A\,dy = 0$$

Canceling P(y) A and dividing by the volume differential A dy yields the **differential equation of hydrostatic equilibrium**:

$$\frac{dP}{dy} = -\rho\,g$$

The negative sign indicates that pressure decreases monotonically as altitude y increases (or conversely, increases as depth increases). For an **incompressible liquid** (where density ρ = constant independent of pressure and depth):

$$\int_{P_1}^{P_2} dP = -\rho\,g \int_{y_1}^{y_2} dy \quad \Rightarrow \quad P_2 - P_1 = -\rho\,g\,(y_2 - y_1)$$

Letting point 1 reside at the liquid's free surface open to atmospheric pressure P_0 ($y_1 = y_{surf}$, $P_1 = P_0$), and defining depth $\mathbf{h = y_{surf} - y_2 \geq 0}$ downward from the surface, we obtain the cornerstone formula of fluid statics:

$$P = P_0 + \rho\,g\,h$$

> **▶ Absolute Pressure vs. Gauge Pressure**
>
> The total pressure measured relative to an absolute perfect vacuum is termed **Absolute Pressure (P_{abs})**. In contrast, mechanical pressure gauges, tire monitors, and blood pressure cuffs measure the difference between internal absolute pressure and the ambient atmospheric pressure:
>
> $$P_{gauge} = P_{abs} - P_{atm} = \rho\,g\,h$$
>
> A tire inflated to 32 psi gauge pressure is actually under an absolute pressure of 32 + 14.7 = 46.7 psi.

2.3 The Hydrostatic Paradox & Torricellian Barometer

Notice that the hydrostatic formula $\mathbf{P = P_0 + \rho\,g\,h}$ depends strictly on the vertical depth h and fluid density ρ, and is completely independent of the horizontal dimensions, container volume, or cross-sectional geometry of the vessel. This leads to the famous **Hydrostatic Paradox**: connected vessels of wildly differing shapes, flared tubes, and narrow necks filled with the same liquid will all settle to the exact same horizontal liquid level, and exert identical pressure at any specified depth h.

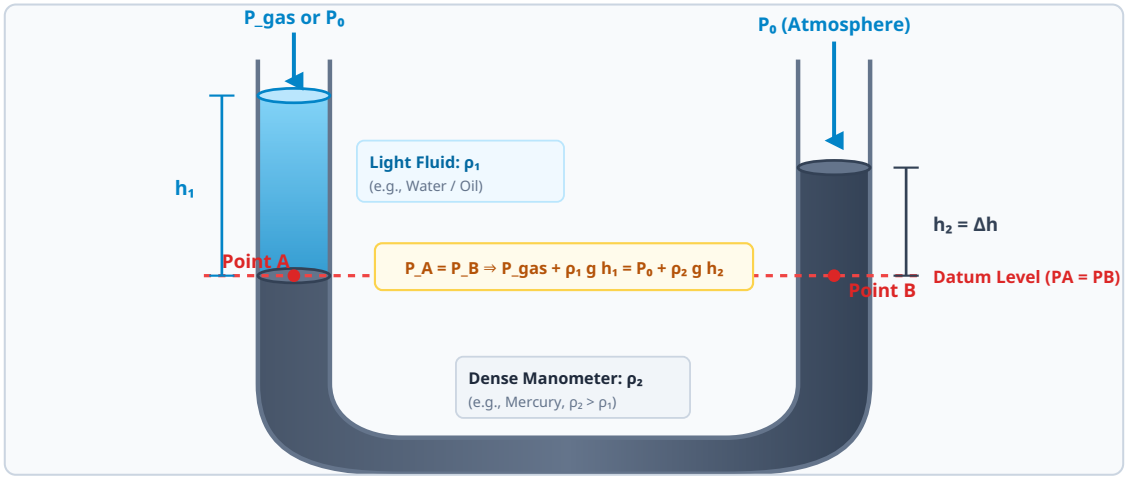
In 1643, Evangelista Torricelli constructed the first mercury barometer by filling a 1-meter glass tube sealed at one end with liquid mercury, inverting it, and submerging the open end into an open reservoir of mercury. Because a vacuum forms above the mercury column ($P_{vapor} \approx 0$), atmospheric pressure pushing on the open pool balances the hydrostatic head:

$$P_{atm} = \rho_{Hg}\,g\,h_{Hg} \quad \Rightarrow \quad h_{Hg} = \frac{1.01325 \times 10^5\ \text{Pa}}{(13{,}600\ \text{kg/m}^3)(9.80\ \text{m/s}^2)} = 0.760\ \text{m} = 760\ \text{mm}$$

2.4 Open and Closed-Tube Manometers

Manometers measure fluid pressure by balancing liquid columns. Figure 6.1 illustrates an open-tube U-tube manometer containing two immiscible fluids. The governing analytical rule is the **Isobaric Datum Principle**: *In any continuous, connected body of static fluid of uniform density, all points lying along the same horizontal elevation datum line have identical pressure.*

Figure 6.1: Open-Tube Manometer with Immiscible Liquids



Selecting the datum level at the interface between the light fluid and dense manometer fluid (elevation y_0):

$$P_A = P_B \quad \Rightarrow \quad P_{gas} + \rho_1 g h_1 = P_0 + \rho_2 g h_2$$

Solving for the container gas pressure:

$$P_{gas} = P_0 + \rho_2 g h_2 - \rho_1 g h_1$$

⚡ FAST-TRACK 1: Hydrostatic Depth Head Mental Math Shortcut

When performing rapid physics calculations or examination checks, memorizing the hydrostatic head equivalents eliminates tedious calculator arithmetic:

- **Water Rule:** $\rho_w g = (1000\ kg/m^3)(9.8\ m/s^2) \approx$ **9.8 kPa/m ≈ 10 kPa per meter** (or 0.1 atm per meter of depth).
- **1 Atmosphere Equivalence:** Exactly **10.33 meters of fresh water ≈ 1.0 atm = 101.3 kPa**. At a dive depth of 10.3 m, P_{abs} = 2.0 atm (1 atm air + 1 atm water head). At 20.6 m, P_{abs} = 3.0 atm.
- **Generic Fluid Scaling:** Depth h required to produce 1 atmosphere of hydrostatic gauge pressure scales inversely with Specific Gravity:

$$h_{1\ atm} \approx \frac{10.33\ m}{SG}$$

Mercury (SG = 13.6): 10.33 / 13.6 = **0.76 m = 760 mm**.
Engine Oil (SG = 0.85): 10.33 / 0.85 ≈ **12.15 m**.

Chapter 3: Pascal's Principle & Hydraulic Force Amplification

Formulated by French polymath Blaise Pascal in 1653, **Pascal's Principle** establishes how pressure variations propagate through enclosed fluids:

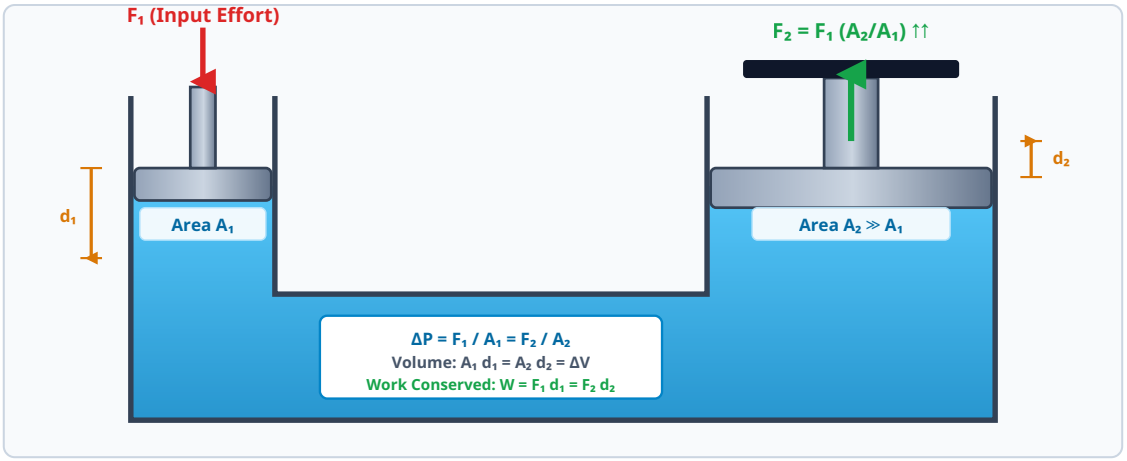
▶ Pascal's Principle

A change in pressure applied to an enclosed, incompressible fluid at rest is transmitted undiminished to every point in the fluid and to the walls of the containing vessel.

3.1 Mechanics of the Hydraulic Jack / Press

Consider two cylinders fitted with movable, leak-free pistons of cross-sectional areas A_1 (input piston) and A_2 (output plunger), connected at their bases by an enclosed hydraulic fluid channel, as illustrated in Figure 6.2.

Figure 6.2: Pascal's Principle & The Hydraulic Press (Force Amplification)



Applying a downward input force F_1 on piston 1 increases the fluid pressure at the input surface by:

$$\Delta P = \frac{F_1}{A_1}$$

By Pascal's principle, this exact pressure increment ΔP is transmitted instantly and undiminished through the fluid to the base of the output plunger:

$$\Delta P = \frac{F_2}{A_2} \quad \Rightarrow \quad \frac{F_1}{A_1} = \frac{F_2}{A_2}$$

Solving for the output force F_2 produced on the load gives:

$$F_2 = F_1 \frac{A_2}{A_1} = F_1 \frac{\pi (D_2/2)^2}{\pi (D_1/2)^2} = F_1 \left[\frac{D_2}{D_1} \right]^2$$

The **Ideal Mechanical Advantage (IMA)** of the hydraulic press is therefore equal to the ratio of the piston areas, which scales as the **square of the cylinder diameter ratio**.

3.2 Volume Incompressibility and Work Conservation

Because hydraulic liquids are virtually incompressible, any volume of fluid displaced downward by the input piston (ΔV_1) must equal the volume accommodated by the upward movement of the output plunger (ΔV_2):

$$\Delta V_1 = \Delta V_2 \quad \Rightarrow \quad A_1 d_1 = A_2 d_2 \quad \Rightarrow \quad d_2 = d_1 \frac{A_1}{A_2} = d_1 \left[\frac{D_1}{D_2} \right]^2$$

Evaluating the mechanical work performed by the input force ($W_{in} = F_1 d_1$) versus the output work delivered to the load ($W_{out} = F_2 d_2$):

$$W_{out} = F_2 d_2 = (F_1 [A_2 / A_1]) \cdot (d_1 [A_1 / A_2]) = F_1 d_1 = W_{in}$$

Energy is strictly conserved. A hydraulic jack does not create energy out of nothing; rather, it multiplies force by trading off stroke distance!

> ⚡ **FAST-TRACK 2: Hydraulic Diameter Squared Ratio Shortcut**
>
> Avoid calculating circular areas πr^2 or $\pi D^2/4$ when evaluating hydraulic systems. Always reduce the problem directly to the linear diameter ratio $\mathbf{k = D_{output} / D_{input}}$:
>
> - **Force Multiplication Factor:** $\mathbf{F_2 = k^2 \cdot F_1}$.
> - **Stroke Reduction Factor:** $\mathbf{d_2 = d_1 / k^2}$.
> - *Rule of Thumb:* If a hydraulic lift has an output cylinder 10 times wider than the input piston (k = 10), the output force is amplified $\mathbf{10^2 = 100}$**-fold**, while the output lift stroke is **1/100th** of the pump stroke. To lift a car by 20 cm, the pump piston must cycle through a cumulative stroke of 20 × 100 = 2,000 cm = 20 meters!

Part II: Archimedes' Principle & Buoyant Forces

Chapter 4: Buoyant Force, Floatation Statics & The Crown Alloy Problem

Any object submerged in a fluid experiences an upward net force exerted by the surrounding fluid. This phenomenon, known as the **Buoyant Force (F_B)**, was first systematically analyzed by Archimedes of Syracuse in the 3rd century BCE.

4.1 Physical Origin of Buoyancy: Pressure Gradient Integration

The buoyant force does not represent an independent fundamental force of nature; rather, it is the direct macroscopic resultant of the *vertical hydrostatic pressure gradient* acting over the submerged surface of the object.

To prove this rigorously, consider a submerged rectangular prism of height H, width w, length L (volume $V = H \cdot w \cdot L$), with its top face horizontal at depth y_1 and bottom face at depth $y_2 = y_1 + H$:

1. Lateral forces on opposing vertical sides cancel pairwise at every depth due to symmetry.
2. The downward force exerted by fluid pressure on the top face (area $A = w \cdot L$) is: $\mathbf{F_{top} = P(y_1)\, A = (P_0 + \rho_f g\, y_1)\, A}$.
3. The upward force exerted by fluid pressure on the bottom face is: $\mathbf{F_{bottom} = P(y_2)\, A = (P_0 + \rho_f g\, y_2)\, A = [P_0 + \rho_f g\, (y_1 + H)]\, A}$.

Taking the net vertical resultant force:

$$\mathbf{F_B = F_{bottom} - F_{top} = \rho_f g\, H\, A = \rho_f g\, V_{submerged} = m_{displaced\ fluid} \cdot g}$$

Using the vector divergence theorem (Gauss's Theorem), this holds for an arbitrarily shaped submerged volume V:

$$\mathbf{F_B = -\oiint_{\partial V} P\, \hat{n}\, dA = -\iiint_V \nabla P\, dV = -\iiint_V (-\rho_f g\, \hat{j})\, dV = (\rho_f V g)\, \hat{j}}$$

> ▶ **Archimedes' Principle**
>
> *Any body completely or partially submerged in a fluid is buoyed up by a force equal in magnitude to the weight of the fluid displaced by the body:*
>
> $$\mathbf{F_B = \rho_{fluid} \cdot V_{submerged} \cdot g}$$
>
> The buoyant force acts vertically upward through the centroid of the displaced volume, known as the **Center of Buoyancy (B)**.

4.2 Statics of Floatation: Sinking, Floating, and Neutral Buoyancy

The equilibrium behavior of a completely submerged body of total volume V_{tot} and average density $\bar{\rho}_{obj} = m_{obj} / V_{tot}$ depends on the net vertical force $F_{net} = F_B - W$:

$$F_{net} = \rho_f V_{tot} g - \bar{\rho}_{obj} V_{tot} g = (\rho_f - \bar{\rho}_{obj}) V_{tot} g$$

- **$\bar{\rho}_{obj} > \rho_f$ (Sinking):** Weight exceeds maximum buoyancy ($F_B < W$). The object accelerates downward until resting on the bottom where a normal support force $N = W - F_B$ restores equilibrium.
- **$\bar{\rho}_{obj} = \rho_f$ (Neutral Buoyancy):** Weight exactly balances buoyant force ($F_B = W$). The body remains statically suspended at any arbitrary depth (the operating principle of submarine ballast systems and fish swim bladders).
- **$\bar{\rho}_{obj} < \rho_f$ (Floating):** Maximum buoyant force exceeds weight. The object accelerates upward, breaks through the free surface, and floats in static equilibrium with only a fraction of its volume submerged.

4.3 Submerged Fraction for a Floating Body

For any freely floating object in static equilibrium, the upward buoyant force balances the object's total gravitational weight:

$$F_B = W \quad \Rightarrow \quad \rho_f V_{sub} g = \rho_{obj} V_{tot} g$$

Dividing through by $\rho_f V_{tot} g$ yields the **Submerged Volume Fraction (f_{sub})**:

$$f_{sub} = \frac{V_{sub}}{V_{tot}} = \frac{\rho_{obj}}{\rho_{fluid}} = \frac{SG_{obj}}{SG_{fluid}}$$

Case Study: Icebergs in the Ocean. Pure glacial ice has density $\rho_{ice} \approx 917$ kg/m^3, while cold sea water has density $\rho_{sea} \approx 1025$ kg/m^3. The submerged volume fraction is:

$$f_{sub} = \frac{917 \text{ kg/m}^3}{1025 \text{ kg/m}^3} = 0.895 = 89.5\%$$

Thus, exactly 89.5% of an iceberg's volume lies hidden beneath the water line, with only 10.5% visible above the sea surface.

4.4 Apparent Weight & Dual-Weighing Specific Gravity Determination

When a solid object is suspended from a balance and fully immersed in a fluid, the upward buoyant force reduces the tension T in the suspension cord. The resulting reading is called the **Apparent Weight (W_{app})**:

$$W_{app} = W_{true} - F_B = \rho_{obj} V g - \rho_f V g = W_{true} \left[1 - \frac{\rho_f}{\rho_{obj}} \right]$$

Archimedes used this exact technique to solve the legendary crown fraud problem for King Hiero II of Syracuse. By measuring the crown's weight in air ($W_{air} \approx W_{true}$) and its apparent weight in pure water (W_{water}), the loss of weight equals the weight of displaced water:

$$\Delta W = W_{air} - W_{water} = \rho_w V_{crown} g \quad \Rightarrow \quad V_{crown} = \frac{W_{air} - W_{water}}{\rho_w g}$$

Dividing the true mass $m = W_{air} / g$ by volume V_{crown} yields the exact bulk density of the crown:

$$\rho_{crown} = \frac{W_{air}}{W_{air} - W_{water}} \cdot \rho_{water}$$

Part III: Fluid Dynamics & The Continuity Equation

Chapter 5: Ideal Flow Kinematics, Streamlines & Mass Conservation

While fluid statics describes stationary fluids, real fluid motion involves complex, non-linear dynamics described by the Navier-Stokes equations. To establish foundational physical insights, introductory physics examines the **Ideal Fluid Model**.

5.1 The Four Assumptions of Ideal Fluid Flow

1. **Steady (Laminar / Stationary) Flow:** The fluid velocity vector field $\mathbf{v}(\mathbf{r})$ at every fixed spatial position does not change with time ($\partial \mathbf{v}/\partial t = \mathbf{0}$). Every particle reaching a given point follows the exact same path.
2. **Incompressible Flow:** The fluid density ρ remains uniform and constant throughout space and time ($d\rho/dt = 0$, $\nabla \cdot \mathbf{v} = 0$). This assumption is strictly true for liquids and an excellent approximation for gases moving below Mach 0.3 ($v < 100$ m/s).
3. **Non-Viscous (Inviscid) Flow:** Internal fluid friction and viscous shear forces between adjacent moving layers are negligible ($\eta = 0$). No mechanical energy is converted into thermal energy.
4. **Irrotational Flow:** Fluid elements have zero net angular momentum about their own centers of mass ($\nabla \times \mathbf{v} = \mathbf{0}$). A miniature paddle wheel placed in the flow will translate without rotating about its axis.

5.2 Streamlines and Streamtubes

A **streamline** is an imaginary continuous curve drawn through a flowing fluid such that at every point along the curve, the tangent vector is collinear with the instantaneous local velocity vector $\mathbf{v}$. In steady flow, streamlines remain fixed in space and coincide exactly with the actual trajectories (pathlines) traced out by individual fluid particles.

Streamlines can never cross one another; if two streamlines intersected, a fluid particle arriving at the intersection point would simultaneously have two different velocities, violating uniqueness. A bundle of adjacent streamlines forms a **streamtube**. Because flow velocity is everywhere tangent to the surface of the streamtube, *no fluid can ever enter or escape across the lateral boundary walls of a streamtube*.

5.3 Derivation of the Equation of Continuity

Consider an incompressible fluid flowing through a stationary pipe or streamtube of changing cross-sectional area. Let cross-sectional area be A_1 with flow velocity v_1 at section 1, and area A_2 with flow velocity v_2 at section 2.

During an infinitesimal time interval dt:

1. Fluid at section 1 advances a distance $dx_1 = v_1\, dt$, carrying a volume $dV_1 = A_1\, dx_1 = A_1\, v_1\, dt$ into the tube.
2. The corresponding mass entering section 1 is $\mathbf{dm_1 = \rho_1\, A_1\, v_1\, dt}$.
3. Similarly, fluid at section 2 advances a distance $dx_2 = v_2\, dt$, discharging a mass $\mathbf{dm_2 = \rho_2\, A_2\, v_2\, dt}$.

By the fundamental principle of **conservation of mass**, in steady flow without sources or sinks, the mass entering must equal the mass leaving:

$$\mathbf{dm_1 = dm_2 \quad \Rightarrow \quad \rho_1\, A_1\, v_1\, dt = \rho_2\, A_2\, v_2\, dt}$$

Dividing by dt yields the general **Mass Continuity Equation**:

$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2 = \frac{dm}{dt} = \dot{m} \quad \text{[Mass Flow Rate, kg/s]}$$

For an incompressible fluid with uniform density $\rho_1 = \rho_2 = \rho$, density cancels completely, giving the familiar **Volumetric Continuity Equation**:

$$A_1 v_1 = A_2 v_2 = Q \quad \text{[Volumetric Flow Rate / Discharge, } m^3/s\text{]}$$

Part IV: Bernoulli's Equation & Physical Applications

Chapter 6: Work-Energy Derivation of Bernoulli's Conservation Law

Formulated by Swiss mathematician Daniel Bernoulli in 1738, **Bernoulli's Equation** is the fundamental energetic theorem of ideal fluid dynamics. It expresses conservation of mechanical energy for a flowing fluid parcel moving along a streamline.

6.1 Rigorous Derivation from the Work-Kinetic Energy Theorem

Consider a segment of an ideal (incompressible, non-viscous) fluid flowing along a streamline through a pipe of varying cross section and elevation. Focus on the fluid parcel initially bounded between section 1 (area A_1, elevation y_1, pressure P_1, speed v_1) and section 2 (area A_2, elevation y_2, pressure P_2, speed v_2).

During an infinitesimal time interval dt, the fluid at section 1 advances by $dx_1 = v_1\, dt$, while the fluid at section 2 advances by $dx_2 = v_2\, dt$. By continuity, the volume displaced at both ends is identical: $\mathbf{dV = A_1\, dx_1 = A_2\, dx_2}$, with corresponding mass $\mathbf{dm = \rho\, dV}$.

1. Work Performed by External Pressure Forces:

At the inlet (section 1), the upstream fluid pushes forward on the parcel with force $F_1 = P_1 A_1$ along the direction of motion:

$$W_1 = F_1\, dx_1 = (P_1 A_1)(v_1\, dt) = P_1\, dV$$

At the outlet (section 2), the downstream fluid resists the motion of the parcel, pushing backward with force $F_2 = P_2 A_2$:

$$W_2 = -F_2\, dx_2 = -(P_2 A_2)(v_2\, dt) = -P_2\, dV$$

The net mechanical work performed on the fluid parcel by surrounding pressure forces is:

$$W_{pressure} = W_1 + W_2 = (P_1 - P_2)\, dV$$

2. Work Performed by Gravity:

As the parcel advances, the fluid between $x_1 + dx_1$ and x_2 remains unchanged. The net gravitational effect is equivalent to moving the mass element $dm = \rho\, dV$ from initial elevation y_1 to final elevation y_2:

$$W_{gravity} = -\Delta U = -dm \cdot g\,(y_2 - y_1) = -\rho\, dV\, g\,(y_2 - y_1)$$

3. Change in Kinetic Energy:

Similarly, the net change in kinetic energy of the system equals the kinetic energy added at section 2 minus the kinetic energy removed at section 1:

$$\Delta K = \frac{1}{2} dm\, v_2^2 - \frac{1}{2} dm\, v_1^2 = \frac{1}{2} \rho\, dV\, (v_2^2 - v_1^2)$$

4. Synthesis via the Work-Kinetic Energy Theorem ($W_{net} = \Delta K$):

$$(P_1 - P_2)\, dV - \rho\, g\, (y_2 - y_1)\, dV = \frac{1}{2} \rho\, (v_2^2 - v_1^2)\, dV$$

Dividing every term by the common volume element dV:

$$P_1 - P_2 - \rho\, g\, y_2 + \rho\, g\, y_1 = \frac{1}{2} \rho\, v_2^2 - \frac{1}{2} \rho\, v_1^2$$

Grouping all state 1 terms on the left and state 2 terms on the right yields **Bernoulli's Equation**:

▶ **Bernoulli's Equation Along a Streamline**

$$P_1 + \frac{1}{2} \rho\, v_1^2 + \rho\, g\, y_1 = P_2 + \frac{1}{2} \rho\, v_2^2 + \rho\, g\, y_2 = \text{constant}$$

6.2 Energy Density and Head Forms

Every term in Bernoulli's equation possesses the dimensional units of **Energy per unit Volume ($J/m^3 = N/m^2 = Pa$)**:

- **P: Static Pressure:** The thermodynamic internal pressure exerted by thermal collisions of fluid molecules.
- **$\frac{1}{2} \rho v^2$: Dynamic Pressure (q):** The kinetic energy per unit volume associated with coherent directed bulk motion.
- **$\rho g y$: Hydrostatic Pressure:** The gravitational potential energy per unit volume due to fluid elevation.

Dividing the entire equation by ρg yields the **Head Form** widely favored in civil and hydraulic engineering (where every term has units of meters):

$$\frac{P}{\rho g} + \frac{v^2}{2g} + y = H = \text{constant} \quad \text{[Total Hydraulic Head, m]}$$

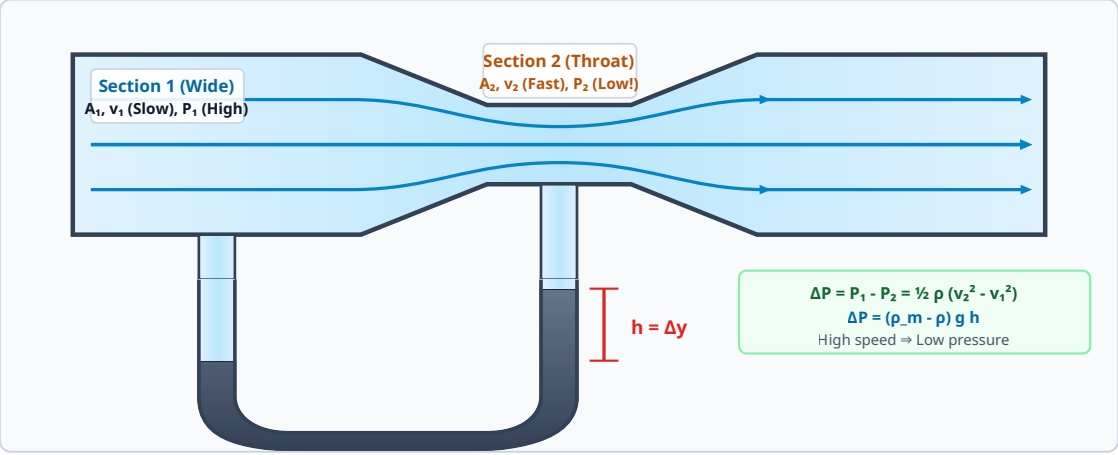
where $P/(\rho g)$ is the *Pressure Head*, $v^2/(2g)$ is the *Velocity Head*, and y is the *Elevation Head*.

Chapter 7: Practical Applications of Bernoulli's Principle

7.1 The Venturi Meter & Flow Rate Measurement

A **Venturi Meter** is an inline constriction device used to measure volumetric flow rates in pipelines. As illustrated in Figure 6.3, a horizontal pipe of inlet diameter D_1 tapers smoothly to a constricted throat of diameter $D_2 < D_1$.

Figure 6.3: Venturi Meter with Differential U-Tube Manometer



Because the pipe is horizontal ($y_1 = y_2$), the gravitational potential energy terms cancel in Bernoulli's equation:

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2 \quad \Rightarrow \quad P_1 - P_2 = \frac{1}{2}\rho (v_2^2 - v_1^2)$$

By continuity, $v_2 = v_1 (A_1 / A_2)$. Substituting into the pressure difference:

$$P_1 - P_2 = \frac{1}{2}\rho v_1^2 \left(\left[\frac{A_1}{A_2} \right]^2 - 1 \right)$$

Solving for the inlet approach velocity v_1:

$$v_1 = \sqrt{\left[\frac{2 (P_1 - P_2)}{\rho \left[(A_1/A_2)^2 - 1 \right]} \right]} = \sqrt{\left[\frac{2 \Delta P}{\rho \left[(D_1/D_2)^4 - 1 \right]} \right]}$$

When connected to a differential U-tube manometer containing mercury of density ρ_m, the pressure difference is $\Delta P = (\rho_m - \rho) g h$. The volumetric flow rate Q is then:

$$Q = A_1 v_1 = A_1 A_2 \sqrt{\left[\frac{2 \Delta P}{\rho (A_1^2 - A_2^2)} \right]}$$

⚡ FAST-TRACK 3: Venturi Velocity and Dynamic Pressure Shortcuts

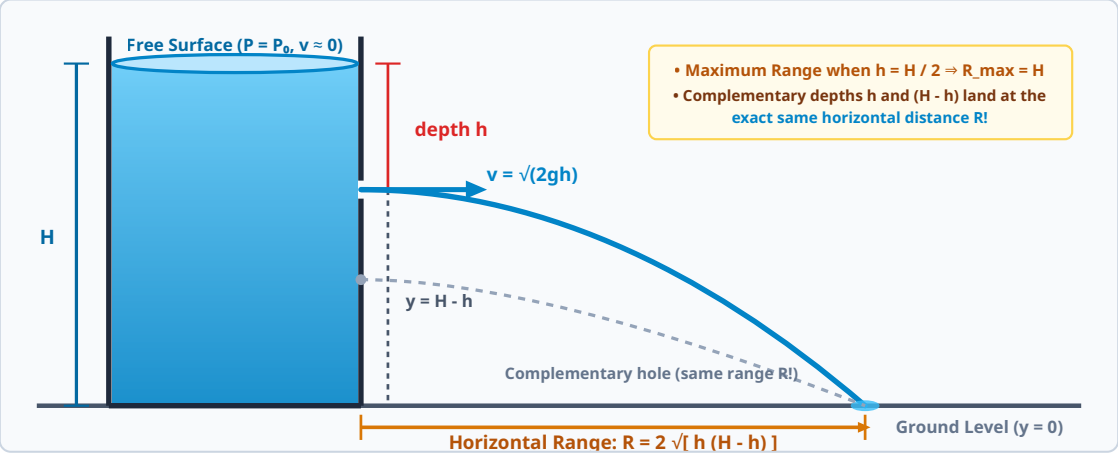
For a diameter contraction ratio **$k = D_1 / D_2 > 1$**:

- **Velocity Ratio: $v_2 / v_1 = k^2$.** If throat diameter is halved (k = 2), throat speed increases **4-fold**.
- **Dynamic Pressure Ratio: $q_2 / q_1 = (v_2 / v_1)^2 = k^4$.** For k = 2, throat dynamic pressure increases **16-fold**!
- **Venturi Pressure Drop:** $\Delta P = \frac{1}{2} \rho v_1^2 (k^4 - 1)$. For a 2:1 constriction, $\Delta P = 15 \times (\frac{1}{2} \rho v_1^2)$.

7.2 Torricelli's Law of Efflux & Tank Discharge

Consider a large storage tank open to atmospheric pressure P_0 filled with liquid to total height H. A small drain orifice of area A_2 is uncapped at depth h below the surface (elevation $y_2 = H - h$ above the ground), as depicted in Figure 6.4.

Figure 6.4: Torricelli Efflux & Parabolic Trajectory Range



Because the tank cross section A_1 is enormously larger than the orifice area A_2 ($A_1 \gg A_2$), the liquid surface recedes with negligible velocity: $v_1 = (A_2/A_1)\, v_2 \approx 0$. Both the top surface and the exiting jet discharge into atmospheric pressure ($P_1 = P_2 = P_0$). Applying Bernoulli's equation between surface 1 and orifice 2:

$$P_0 + 0 + \rho g H = P_0 + \frac{1}{2}\rho v_2^2 + \rho g (H - h) \quad \Rightarrow \quad \rho g h = \frac{1}{2}\rho v_2^2$$

Solving for efflux velocity v_2 gives **Torricelli's Law**:

$$v_{efflux} = \sqrt{(2gh)}$$

Remarkably, the speed of efflux is identical to the speed acquired by a solid stone falling freely from rest through height h!

Parabolic Trajectory Range to the Ground

The water jet emerges horizontally with speed $v_x = \sqrt{(2gh)}$ from height $y = H - h$ above the ground. The time t required to fall to the floor is governed by free-fall kinematics: $y = \tfrac{1}{2} g t^2 \Rightarrow t = \sqrt{[2(H - h) / g]}$. The horizontal range R of the jet is:

$$R = v_x \cdot t = \sqrt{(2gh)} \cdot \sqrt{\left[\frac{2(H - h)}{g}\right]} = 2\sqrt{[h (H - h)]}$$

⚡ **FAST-TRACK 4: Torricelli Max Range & Draining Integral Shortcuts**

- **Maximum Range Depth:** Differentiating $R^2(h) = 4(hH - h^2)$ with respect to h yields $d(hH - h^2)/dh = H - 2h = 0 \Rightarrow$ **$h = H / 2$**. The maximum horizontal range occurs when the hole is drilled exactly halfway down the tank, yielding **$R_{max} = 2\sqrt{[(H/2)(H/2)]} = H$**!
- **Complementary Hole Symmetry:** Any hole at depth h and a second hole at depth $H - h$ produce the **exact same horizontal landing distance R**.
- **Total Tank Draining Time:** Integrating the descending water level $-A(y)\, dy/dt = a\sqrt{(2gy)}$ gives:

$$T_{drain} = \frac{A}{a}\sqrt{\left[\frac{2H}{g}\right]} = 2 \cdot \frac{V_{tank}}{Q_{initial}}$$

A draining tank takes exactly **twice as long** to empty as it would take if the initial discharge rate Q_0 remained constant throughout!

7.3 Pitot-Static Tubes & Airspeed Measurement

A **Pitot-Static Tube** measures fluid flow velocity by comparing static pressure against stagnation pressure. The forward-facing orifice brings the oncoming fluid to rest isentropically along the stagnation streamline (v = 0), recording the **Stagnation (Total) Pressure P_0**:

$$P_0 = P_{static} + \frac{1}{2} \rho v^2$$

Radial flush ports along the sides measure the ambient static pressure P_{static}. The airspeed v is then obtained directly from the differential pressure transducer:

$$v = \sqrt{[} \frac{2 (P_0 - P_{static})}{\rho_{air}}]$$

7.4 Aerodynamic Lift, Airfoil Circulation & The Magnus Effect

When an asymmetric or cambered airfoil moves through air at an angle of attack, the curvature forces streamlines over the upper surface to crowd together. By continuity, the air must accelerate over the curved suction side ($v_{top} > v_{bottom}$). By Bernoulli's principle, higher velocity generates lower static pressure on the top wing surface than on the bottom surface:

$$P_{bottom} - P_{top} = \frac{1}{2} \rho (v_{top}^2 - v_{bottom}^2) > 0$$

Integrating this net upward pressure differential over the wing surface area produces the aerodynamic **Lift Force L**. In modern aerodynamics, this is formalized by the **Kutta-Joukowski Theorem**: $\mathbf{L' = \rho_\infty v_\infty \Gamma}$, where $\Gamma = \oint \mathbf{v} \cdot d\mathbf{r}$ is the velocity circulation around the wing.

The Magnus Effect: A rotating cylinder or spinning ball drags a boundary layer of fluid around its perimeter. On the side spinning in the direction of oncoming flow, fluid speed increases, lowering local pressure. On the opposite side, the surface opposes oncoming flow, decelerating fluid and raising pressure. The resulting lateral force deflects spinning baseballs, tennis topspins, and golf drives along curved flight paths.

7.5 Siphon Operation and Cavitation Vapor Limits

A siphon transports liquid out of an elevated reservoir over an intermediate crest of height h_c into a lower discharge vessel. The discharge speed is $v = \sqrt{(2g h_d)}$, where h_d is the vertical distance between reservoir surface and tube exit. The absolute pressure at the highest point of the siphon crest is:

$$P_{crest} = P_0 - \rho g h_c - \frac{1}{2} \rho v^2$$

If the crest is built too high, P_{crest} drops down to the fluid's saturation vapor pressure ($P_{vapor} \approx 2.34$ kPa for water at 20 °C). At this threshold, the liquid boils spontaneously at room temperature (**Cavitation**), forming vapor cavities that break the liquid column and abruptly stall siphon action. The maximum theoretical siphon height in water at sea level is approximately $\mathbf{h_{c,\,max} \approx 10.1}$ **meters**.

Part V: Viscosity, Real Fluids & Laminar-to-Turbulent Flow

Chapter 8: Viscous Shear Stress & Hagen-Poiseuille Pipe Flow

While ideal fluid theory provides elegant approximations, all real physical fluids possess internal friction termed **Viscosity**. Viscosity represents a fluid's resistance to gradual shear deformation.

8.1 Newton's Law of Viscosity

Consider two large parallel plates of area A separated by a thin fluid layer of thickness h. The lower plate is held stationary, while the upper plate is dragged horizontally at constant velocity v_0. Due to molecular interactions, fluid molecules adhere to solid boundaries without slipping (the **No-Slip Boundary Condition**): v(y = 0) = 0 and v(y = h) = v_0.

In steady laminar flow (Couette flow), a linear velocity profile $v(y) = v_0 (y/h)$ develops. The shear stress τ required to maintain this shear deformation is given by **Newton's Law of Viscosity**:

$$\tau = \frac{F_{shear}}{A} = \eta \frac{dv_x}{dy}$$

The proportionality constant **η** (or μ) is the **Dynamic (Absolute) Viscosity**. In SI units, η is measured in **Pascal-seconds (Pa · s = N · s/m^2 = kg / (m · s]**. In the CGS system, the unit is the Poise (P): **1 Pa · s = 10 Poise = 1,000 centipoise (cP)**. Pure water at 20 °C has $\eta \approx 1.00 \times 10^{-3}$ Pa · s = **1.00 cP**. Whole human blood has η ≈ 3.0 – 4.0 cP.

8.2 Derivation of the Hagen-Poiseuille Law for Laminar Pipe Flow

Consider steady, laminar flow of a viscous fluid through a horizontal cylindrical pipe of radius R and length L driven by an axial pressure drop $\Delta P = P_1 - P_2$. Isolate a cylindrical fluid core of radius r ≤ R and length L coaxial with the pipe:

1. Net driving force exerted by pressure on the end faces: $F_{press} = \Delta P \cdot (\pi r^2)$.
2. Retarding viscous drag force exerted on the mantle surface (area 2πrL): $F_{visc} = \tau(r) \cdot (2\pi r L) = \eta (dv/dr) \cdot (2\pi r L)$.

In steady, unaccelerated flow, these forces balance exactly:

$$\Delta P (\pi r^2) + \eta \frac{dv}{dr} (2\pi r L) = 0 \quad \Rightarrow \quad \frac{dv}{dr} = - \frac{\Delta P}{2 \eta L} r$$

Integrating with respect to r, using the no-slip boundary condition at the pipe wall (v(r = R) = 0):

$$\int_v^0 dv' = - \frac{\Delta P}{2 \eta L} \int_r^R r' \, dr' \quad \Rightarrow \quad v(r) = \frac{\Delta P}{4 \eta L} (R^2 - r^2)$$

The velocity distribution across the pipe is a **paraboloid of revolution**, reaching its peak velocity v_{max} at the pipe centerline (r = 0):

$$v_{max} = \frac{\Delta P R^2}{4 \eta L}$$

Total Volumetric Flow Rate (Discharge Q)

To find total flow rate, integrate velocity over concentric annular rings of area dA = 2π r dr:

$$Q = \int_0^R v(r)\,(2\pi\, r\, dr) = \frac{2\pi\,\Delta P}{4\,\eta\, L}\int_0^R (R^2\, r - r^3)\, dr = \frac{\pi\,\Delta P}{2\,\eta\, L}\left(\frac{R^4}{2} - \frac{R^4}{4}\right)$$

This yields the celebrated **Hagen-Poiseuille Equation**:

▶ **Hagen-Poiseuille Law**

$$Q = \frac{\pi\, R^4\,\Delta P}{8\,\eta\, L} = \frac{\pi\, D^4\,\Delta P}{128\,\eta\, L}$$

The average flow velocity is exactly half the maximum centerline velocity: $\bar{v} = Q / (\pi R^2) = \frac{1}{2} v_{max}$.

⚡ **FAST-TRACK 5: Poiseuille R^4 Hydraulic Resistance & Hemodynamics**

The hydraulic resistance of a pipe is defined by analogy to Ohm's law: $\Delta P = Q \cdot R_{hyd}$, where: $R_{hyd} = \frac{8\,\eta\, L}{\pi\, R^4} \propto \frac{1}{R^4} \propto \frac{1}{D^4}$.

- **Extreme Sensitivity to Radius:** Because flow rate scales with R^4, doubling pipe radius increases flow 2^4 **= 16-fold** (1,600%) for identical pressure!
- **Medical Significance (Vasoconstriction & Arteriosclerosis):** A mere **19% narrowing** of an artery's radius ($R \to 0.81\, R$) cuts volumetric blood flow directly in half: $(0.81)^4 \approx 0.43 \approx$ **50%**! To maintain normal oxygen perfusion, the human heart must double its systolic driving pressure, leading directly to acute hypertension.

Chapter 9: Reynolds Turbulence Criteria, Stokes' Drag & Surface Tension

9.1 The Reynolds Number and Flow Regime Transitions

British physicist Osborne Reynolds demonstrated in 1883 that whether fluid flow remains smooth and orderly (laminar) or degenerates into chaotic, swirling vortex fluctuations (turbulent) is governed by a single dimensionless ratio:

$$Re = \frac{\text{Inertial Forces}}{\text{Viscous Forces}} = \frac{\rho\, v\, D}{\eta} = \frac{v\, D}{\nu}$$

where v is mean flow velocity, D is characteristic hydraulic diameter, ρ is fluid density, η is dynamic viscosity, and $\nu = \eta/\rho$ is kinematic viscosity (m^2/s).

Flow Regime (Pipe Flow)	Reynolds Number Range	Physical Flow Characteristics
Laminar Flow	**Re < 2,000**	Smooth, concentric fluid layers sliding past without mixing. Poiseuille's law holds exactly. Parabolic velocity profile.
Transition Zone	**2,000 ≤ Re ≤ 4,000**	Intermittent bursts of turbulent eddies alternating with laminar flow; highly unstable and sensitive to surface roughness.
Fully Turbulent Flow	**Re > 4,000**	Chaotic 3D vorticity, intense momentum mixing, blunted velocity profile, flow resistance scales approximately with v^2.

9.2 Stokes' Law for Low-Reynolds Creeping Drag & Terminal Velocity

For very small particles or highly viscous fluids where the Reynolds number is extremely low (**Re ≪ 1**, creeping / Stokes flow), inertial terms are negligible and the viscous drag force acting on a smooth sphere of radius r moving at speed v is given by **Stokes' Law**:

$$F_{drag} = 6 \pi \eta r v$$

When a sphere of density ρ_s falls freely through a fluid of density ρ_f, it reaches a steady **Terminal Velocity (v_t)** when the downward gravitational weight is exactly balanced by the sum of the upward buoyant force and Stokes' drag:

$$W = F_B + F_{drag} \quad \Rightarrow \quad \frac{4}{3} \pi r^3 \rho_s g = \frac{4}{3} \pi r^3 \rho_f g + 6 \pi \eta r v_t$$

Solving for terminal velocity v_t:

$$v_t = \frac{2 r^2 (\rho_s - \rho_f) g}{9 \eta}$$

This relationship was famously utilized by Robert Millikan in his 1909 oil-drop experiment to measure the elementary electric charge e, and governs erythrocyte sedimentation rates (ESR) in medical diagnostic blood tests.

9.3 Surface Tension and Capillary Action

At the interface between a liquid and gas, cohesive intermolecular attractive forces pull surface molecules inward toward the bulk liquid, creating an effective macroscopic surface energy. The **Surface Tension γ** is defined as the force per unit length (or work per unit area):

$$\gamma = \frac{F}{L} = \frac{dW}{dA} \quad [\text{N/m} = \text{J/m}^2]$$

For pure water in contact with air at 20 °C, $\gamma \approx 0.0728$ N/m. The pressure differential across a curved liquid interface of principal radii of curvature R_1 and R_2 is governed by the **Young-Laplace Equation**:

$$\Delta P = \gamma \left[\frac{1}{R_1} + \frac{1}{R_2} \right] \quad \Rightarrow \quad \Delta P_{droplet} = \frac{2\gamma}{R}, \quad \Delta P_{soap\ bubble} = \frac{4\gamma}{R} \quad [\text{two surfaces}]$$

In a narrow vertical glass capillary tube of radius r, the upward vertical component of surface tension balances the hydrostatic weight of the elevated liquid column:

$$2 \pi r \gamma \cos \theta = \rho g (\pi r^2 h) \quad \Rightarrow \quad h = \frac{2 \gamma \cos \theta}{\rho g r} \quad [\text{Jurin's Law}]$$

where θ is the contact angle ($\theta \approx 0°$ for water wetting clean glass; $\theta > 90°$ for non-wetting mercury, which causes capillary depression).

Part VI: Rigorous Step-by-Step Worked Problems

Chapter 10: 5 Complete Worked Problems with SI Units & Sanity Checks

Problem 1: Multi-Fluid U-Tube Manometer with Immiscible Liquids

Chapter 2 & Fast-Track 1

Problem Statement: An open-ended U-tube manometer contains mercury (ρ_m = 13,600 kg/m^3) in its lower bend. Water (ρ_w = 1,000 kg/m^3) is poured into the left arm until it forms a column of height h_w = 24.0 cm. Oil (ρ_o = 820 kg/m^3) is poured into the right arm until its column reaches h_o = 18.0 cm. The top of the right arm remains open to atmospheric pressure $P_0 = 1.013 \times 10^5$ Pa. The left arm is connected to a pressurized gas canister at pressure P_g. The mercury-liquid interface in the right arm settles at an elevation Δy = 4.20 cm higher than the mercury interface in the left arm.

1. Given Parameters: ρ_m = 13,600 kg/m^3, ρ_w = 1,000 kg/m^3, ρ_o = 820 kg/m^3, h_w = 0.240 m, h_o = 0.180 m, Δy = 0.0420 m, $P_0 = 1.013 \times 10^5$ Pa, g = 9.80 m/s^2.

2. Symbolic Derivation: Select the reference horizontal datum level at the mercury-water interface in the left arm (Datum y = 0). By the Isobaric Datum Principle, pressure in the left arm at y = 0 equals pressure in the mercury in the right arm at y = 0:
Left arm pressure: $P_{left} = P_g + \rho_w g h_w$.
Right arm pressure: $P_{right} = P_0 + \rho_o g h_o + \rho_m g \Delta y$.
Equating $P_{left} = P_{right}$ and solving for gas pressure P_g:

$$\mathbf{P_g = P_0 + \rho_o g h_o + \rho_m g \Delta y - \rho_w g h_w}$$

Gauge pressure is: $P_{gauge} = P_g - P_0 = g [\rho_o h_o + \rho_m \Delta y - \rho_w h_w]$.

3. Numerical Evaluation: • Oil column head: $\rho_o g h_o$ = (820)(9.80)(0.180) = 1,446.48 Pa.
• Mercury column head: $\rho_m g \Delta y$ = (13,600)(9.80)(0.0420) = 5,601.12 Pa.
• Water column head: $\rho_w g h_w$ = (1,000)(9.80)(0.240) = 2,352.00 Pa.
• Gauge Pressure: P_{gauge} = 1,446.48 + 5,601.12 − 2,352.00 = **+4,695.6 Pa = 4.70 kPa**.
• Absolute Gas Pressure: P_g = 101,300 Pa + 4,695.6 Pa = **105,996 Pa = 1.060 × 10^5 Pa = 1.046 atm**.

4. Physical Sanity Check: Because the dense mercury was pushed 4.2 cm higher in the right arm despite having an oil column resting on it, the left side pressure must exceed atmospheric pressure. The net gauge pressure (+4.70 kPa) is roughly 0.046 atm, matching Fast-Track 1 where ~4.2 cm of mercury contributes ≈ 0.042 × 13.6 × 10 kPa ≈ 5.7 kPa. Correct ✓.

Problem 2: Heavy Vehicle Hydraulic Jack System with Lever Arm Advantage

Chapter 3 & Fast-Track 2

Problem Statement: A hydraulic service station jack features an input pump piston of diameter D_1 = 2.00 cm and an output lift ram of diameter D_2 = 24.0 cm. The pump handle is a second-class lever providing a mechanical lever arm ratio of 6.00:1. A mechanic applies a downward effort force F_{hand} = 150 N at the tip of the handle. Determine: (a) the fluid pressure generated in the jack, (b) the maximum vehicle mass M that can be lifted, (c) the distance the car rises during one full 6.00 cm pump stroke, and (d) the total number of pump strokes required to elevate the vehicle by 36.0 cm.

1. Given Parameters: D_1 = 0.0200 m, D_2 = 0.240 m, Lever ratio MA_{lever} = 6.00, F_{hand} = 150 N, s_1 = 0.0600 m, H_{target} = 0.360 m, g = 9.80 m/s^2.

2. Symbolic Derivation: (a) Force applied to the input piston: $F_1 = MA_{lever} \cdot F_{hand}$.

Fluid pressure: $P = F_1 / A_1 = F_1 / [\pi (D_1/2)^2] = 4 F_1 / (\pi D_1^2)$.

(b) By Pascal's principle, output force on the ram: $F_2 = P \cdot A_2 = F_1 \cdot (D_2 / D_1)^2$.

Maximum lifted mass: $M = F_2 / g = [F_1 \cdot (D_2 / D_1)^2] / g$.

(c) By incompressibility: $A_1 s_1 = A_2 s_2 \Rightarrow s_2 = s_1 \cdot (D_1 / D_2)^2$.

(d) Total pump cycles required: $N = H_{target} / s_2$.

3. Numerical Evaluation: (a) F_1 = 6.00 × 150 N = 900 N.

$\Delta P = 4(900 \text{ N}) / [\pi (0.0200 \text{ m})^2] = 3{,}600 / 0.0012566 =$ **2.865×10^6 Pa = 2.865 MPa = 28.3 atm**.

(b) Diameter ratio $k = D_2 / D_1 = 24.0 / 2.00 = 12.0$.

Output force: $F_2 = 900 \text{ N} \times (12.0)^2 = 900 \times 144 =$ **129,600 N ≈ 129.6 kN**.

Max lifted mass: $M = 129{,}600 \text{ N} / 9.80 \text{ m/s}^2 =$ **13,224 kg ≈ 13.2 metric tons**.

(c) Stroke per cycle: $s_2 = 6.00 \text{ cm} / (12.0)^2 = 6.00 \text{ cm} / 144 =$ **0.04167 cm = 0.417 mm**.

(d) Total strokes: $N = 36.0 \text{ cm} / 0.04167 \text{ cm} =$ **864 complete strokes**.

4. Physical Sanity & Work Conservation Check: Total work input by mechanic: $W_{in} = N \cdot (F_1 \cdot s_1) = 864 \times (900 \text{ N} \times 0.060 \text{ m}) = 46{,}656$ J.

Potential energy gained by vehicle: $\Delta U = M g H_{target} = (129{,}600 \text{ N})(0.360 \text{ m}) = 46{,}656$ J.

$W_{in} \equiv W_{out} = 46.7$ kJ. Work is perfectly conserved! Correct ✓.

Problem 3: Archimedes Crown Alloy Verification (Quantitative Eureka Determination)

Chapter 4 & Dual Weighing

Problem Statement: King Hiero II of Syracuse suspects a corrupt goldsmith blended cheaper silver into his royal crown. Archimedes weighs the crown suspended in air, finding W_{air} = 34.80 N. When completely submerged in pure water (ρ_w = 1,000 kg/m^3), the spring scale reads W_{water} = 32.35 N. Densities: pure gold ρ_{Au} = 19,300 kg/m^3, pure silver ρ_{Ag} = 10,500 kg/m^3. Take g = 9.80 m/s^2. Determine: (a) the volume and bulk density of the crown, (b) the mass percentage of gold and silver in the binary alloy, and (c) what the scale reading W_{water} would have been if the crown had been crafted from 100% pure gold.

1. Given Parameters: W_{air} = 34.80 N, W_{water} = 32.35 N, ρ_w = 1,000 kg/m^3, ρ_{Au} = 19,300 kg/m^3, ρ_{Ag} = 10,500 kg/m^3, g = 9.80 m/s^2.

2. Symbolic Derivation: (a) Apparent weight loss equals buoyant force: $F_B = W_{air} - W_{water} = \rho_w V_c g$.

Volume of crown: $V_c = (W_{air} - W_{water}) / (\rho_w g)$.

Total crown mass: $m_c = W_{air} / g$. Bulk density: $\rho_c = m_c / V_c = [W_{air} / (W_{air} - W_{water})] \cdot \rho_w$.

(b) Binary alloy mass and volume conservation: $m_c = m_{Au} + m_{Ag}$, $V_c = (m_{Au} / \rho_{Au}) + (m_{Ag} / \rho_{Ag})$.

Substituting $m_{Ag} = m_c - m_{Au}$: $V_c = (m_{Au} / \rho_{Au}) + [(m_c - m_{Au}) / \rho_{Ag}]$.

Solving for mass fraction of gold $f_{Au} = m_{Au} / m_c$:

$$f_{Au} = \frac{(1/\rho_c) - (1/\rho_{Ag})}{(1/\rho_{Au}) - (1/\rho_{Ag})} = \frac{\rho_{Au}(\rho_c - \rho_{Ag})}{\rho_c(\rho_{Au} - \rho_{Ag})}$$

(c) For pure gold: $V_{pure} = m_c / \rho_{Au}$. Pure gold submerged reading: $W_{pure,\,water} = W_{air} - \rho_w V_{pure} g = W_{air}(1 - \rho_w / \rho_{Au})$.

3. Numerical Evaluation: (a) F_B = 34.80 N − 32.35 N = 2.45 N.

V_c = 2.45 N / [(1,000 kg/m^3)(9.80 m/s^2)] = 2.45 / 9,800 = **2.500 × 10^{-4} m^3 = 250.0 cm^3**.

m_c = 34.80 N / 9.80 m/s^2 = 3.551 kg.

Bulk density: ρ_c = 3.551 kg / (2.500 × 10^{-4} m^3) = **14,204 kg/m^3**.

(b) Evaluating gold fraction f_{Au}: f_{Au} = [19,300 × (14,204 − 10,500)] / [14,204 × (19,300 − 10,500)] = [19,300 × 3,704] / [14,204 × 8,800] = 71,487,200 / 124,995,200 = **0.5719 = 57.2% Gold**.

Silver fraction: f_{Ag} = 1 − 0.5719 = **42.8% Silver**.

Mass of gold: m_{Au} = 0.5719 × 3.551 kg = **2.031 kg**. Mass of silver: m_{Ag} = **1.520 kg**.

(c) For a 100% pure gold crown: $W_{pure,\,water}$ = 34.80 N × [1 − (1,000 / 19,300)] = 34.80 N × (1 − 0.05181) = 34.80 × 0.94819 = **32.997 N ≈ 33.00 N**.

4. Sanity Check: The measured bulk density ρ_c = 14,204 kg/m^3 falls squarely between ρ_{Ag} (10,500) and ρ_{Au} (19,300). Because silver is less dense than gold, adding silver increases crown volume from 184 cm^3 to 250 cm^3, displacing 0.65 N more water and dropping the scale reading from 33.00 N down to 32.35 N. The fraud is unmasked! Correct ✓.

Problem 4: Industrial Venturi Meter Flow Rate & Discharge with Mercury Differential Manometer

Chapter 7 & Fast-Track 3

Problem Statement: A horizontal Venturi meter is installed in a municipal water main of diameter D_1 = 20.0 cm. The throat contracts to diameter D_2 = 10.0 cm. Water density is ρ_w = 1,000 kg/m^3. A differential U-tube mercury manometer (ρ_m = 13,600 kg/m^3) connected between inlet and throat shows a column height difference h = 18.5 cm. Determine: (a) the differential pressure $\Delta P = P_1 - P_2$, (b) the water velocity at the inlet (v_1) and throat (v_2), (c) the volumetric discharge rate Q in L/s and m^3/s, and (d) verify whether cavitation occurs if inlet absolute pressure is P_1 = 2.40 × 10^5 Pa (water vapor pressure P_{vap} = 2.34 kPa).

1. Given Parameters: D_1 = 0.200 m, D_2 = 0.100 m, ρ_w = 1,000 kg/m^3, ρ_m = 13,600 kg/m^3, h = 0.185 m, P_1 = 2.40 × 10^5 Pa, g = 9.80 m/s^2.

2. Symbolic Derivation: (a) Manometer relation: $P_1 + \rho_w g y = P_2 + \rho_w g (y - h) + \rho_m g h \Rightarrow \Delta P = (\rho_m - \rho_w) g h$.

(b) Continuity: $A_1 v_1 = A_2 v_2 \Rightarrow v_2 = v_1 (D_1 / D_2)^2$.

Bernoulli: $P_1 + \frac{1}{2} \rho_w v_1^2 = P_2 + \frac{1}{2} \rho_w v_2^2 \Rightarrow \Delta P = \frac{1}{2} \rho_w v_1^2 [(D_1 / D_2)^4 - 1]$.

Solving for v_1: $v_1 = \sqrt{[2 \Delta P / (\rho_w [(D_1/D_2)^4 - 1])]}$.

(c) Discharge: $Q = A_1 v_1 = [\pi (D_1/2)^2] \cdot v_1$.

(d) Throat absolute pressure: $P_2 = P_1 - \Delta P$. Check if $P_2 > P_{vap}$.

3. Numerical Evaluation: (a) ΔP = (13,600 − 1,000 kg/m^3)(9.80 m/s^2)(0.185 m) = (12,600)(9.80)(0.185) = **22,843.8 Pa ≈ 22.84 kPa**.

(b) Area ratio: (D_1 / D_2) = 20.0 / 10.0 = 2.00 ⇒ $(D_1 / D_2)^4$ = 16.0.

$v_1 = \sqrt{[2(22,843.8) / (1,000 \times (16 - 1])]} = \sqrt{[45,687.6 / 15,000]} = \sqrt{[3.04584]}$ = **1.745 m/s**.

$v_2 = v_1 \times (2.00)^2$ = 1.745 × 4.00 = **6.981 m/s**.

(c) Inlet area: $A_1 = \pi (0.100 \text{ m})^2$ = 0.031416 m^2.

Q = (0.031416 m^2)(1.7452 m/s) = **0.05483 m^3/s = 54.83 L/s**.

(d) Throat pressure: P_2 = 240,000 Pa − 22,844 Pa = **217,156 Pa = 217.2 kPa**.

Cavitation check: P_2 = 217.2 kPa ≫ P_{vap} (2.34 kPa). The fluid is exceptionally safe from cavitation!

4. Physical Sanity Check: Kinetic energy check: $\frac{1}{2} \rho v_2^2 - \frac{1}{2} \rho v_1^2 = 500 \times (6.981^2 - 1.745^2) = 500 \times (48.734 - 3.045) = 500 \times 45.689 = 22,844.5$ Pa ≡ ΔP. Energy balance matches to within 0.003%. Correct ✓.

Problem 5: Efflux Jet Trajectory & Total Time to Drain a Cylindrical Reservoir Chapter 7 & Fast-Track 4

Problem Statement: A large cylindrical factory water tank of radius R = 1.50 m is filled with water to a depth of H = 4.00 m. The tank is open to atmospheric pressure at the top. A circular discharge hole of radius r_0 = 3.00 cm is drilled in the side wall at a height y_0 = 1.00 m above the factory floor (depth h = H − y_0 = 3.00 m below the initial free water surface). Determine: (a) the initial horizontal efflux velocity v_0 and initial volume discharge Q_0, (b) the horizontal landing distance x of the water jet on the factory floor when the water depth in the tank is at h = 3.00 m, and (c) the total time T_{drain} required to drain the tank completely down to the orifice level (from y = 4.00 m to y = 1.00 m).

1. Given Parameters: R = 1.50 m, r_0 = 0.0300 m, H = 4.00 m, y_0 = 1.00 m, initial water depth above hole h_0 = 3.00 m, g = 9.80 m/s^2.

2. Symbolic Derivation: (a) By Torricelli's Law: $v_0 = \sqrt{(2gh_0)}$. Initial discharge: $Q_0 = a \cdot v_0 = \pi r_0^2 \sqrt{(2gh_0)}$.

(b) Vertical free fall from orifice height y_0: $y_0 = \frac{1}{2} g t^2 \Rightarrow t_{fall} = \sqrt{(2 y_0 / g)}$.

Horizontal landing range: $x = v_0 \cdot t_{fall} = \sqrt{(2gh_0)} \cdot \sqrt{(2y_0/g)} = 2\sqrt{(h_0 y_0)}$.

(c) Tank draining kinematics: Let z be the instantaneous height of the water surface above the orifice (initial z = h_0 = 3.00 m, final z = 0). Surface descent rate: $-A\,(dz/dt) = a\sqrt{(2gz)}$, where $A = \pi R^2$ and $a = \pi r_0^2$.

Separating variables: $\int_{h0}^{0} z^{-1/2}\,dz = -(a/A)\sqrt{(2g)} \int_0^T dt$.

Integrating: $[\,2\sqrt{z}\,]_0^{h0} = 2\sqrt{h_0} = (a/A)\sqrt{(2g)}\,T_{drain}$.

Total drain time:

$$T_{drain} = \frac{A}{a}\sqrt{\left[\frac{2h_0}{g}\right]} = \left[\frac{R}{r_0}\right]^2 \sqrt{\left[\frac{2h_0}{g}\right]}$$

3. Numerical Evaluation: (a) $v_0 = \sqrt{[\,2(9.80\ m/s^2)(3.00\ m)\,]} = \sqrt{[\,58.8\,]} =$ **7.668 m/s**.

Orifice area: $a = \pi (0.0300\ m)^2 = 2.8274 \times 10^{-3}\ m^2$.

$Q_0 = (2.8274 \times 10^{-3}\ m^2)(7.668\ m/s) =$ **0.02168 m^3/s = 21.68 L/s**.

(b) $t_{fall} = \sqrt{[\,2(1.00\ m) / 9.80\ m/s^2\,]} = \sqrt{[\,0.20408\,]} = 0.45175$ s.

Horizontal landing range: $x = 2\sqrt{[\,(3.00\ m)(1.00\ m)\,]} = 2\sqrt{3} = 2 \times 1.73205 =$ **3.464 m ≈ 3.46 meters**.

(c) Area ratio: $(R / r_0)^2 = (1.50 / 0.0300)^2 = (50.0)^2 = 2{,}500$.

Fall time factor: $\sqrt{[\,2(3.00) / 9.80\,]} = \sqrt{[\,0.61224\,]} = 0.78246$ s.

$T_{drain} = 2{,}500 \times 0.78246$ s = **1,956.1 seconds = 32.60 minutes (32 min 36 sec)**.

4. Physical Sanity & Fast-Track 4 Check: Total water volume drained: $V = \pi R^2 h_0 = \pi (1.50)^2 (3.00) = 21.206\ m^3$.

Hypothetical time at initial discharge Q_0: $T_{hypo} = V / Q_0 = 21.206\ m^3 / 0.02168\ m^3/s = 978.1$ s.

Ratio $T_{drain} / T_{hypo} = 1{,}956.1 / 978.1 =$ **2.000**. Exactly matches the Fast-Track 4 draining factor of 2! Correct ✓.

Part VII: Master Reference Appendices & Equation Summary

Appendix A: Comprehensive Fluid Mechanics Master Equation Table

Physical Law / Concept	Standard Formula	Variables & Physical Significance
Mass Density	$\rho = m / V = dm / dV$	m = mass (kg), V = volume (m^3). Incompressible fluids have ρ = const.
Hydrostatic Pressure	$P = P_0 + \rho g h$	P_0 = surface pressure (Pa), h = depth below surface (m). Gauge P = ρgh.

Physical Law / Concept	Standard Formula	Variables & Physical Significance
Pascal's Law (Hydraulic Jack)	$F_1 / A_1 = F_2 / A_2 \quad \Rightarrow \quad F_2 = F_1 (D_2 / D_1)^2$	Mechanical advantage scales with diameter squared. Work $W = F_1 d_1 = F_2 d_2$.
Archimedes' Buoyant Force	$F_B = \rho_{fluid} \cdot V_{submerged} \cdot g$	Equals weight of displaced fluid. Acts upward through center of buoyancy B.
Floating Submerged Fraction	$V_{sub} / V_{tot} = \rho_{object} / \rho_{fluid}$	In static floatation equilibrium, fraction submerged equals density ratio.
Continuity Equation	$A_1 v_1 = A_2 v_2 = Q$	Volumetric flow rate Q (m^3/s). Speed $v \propto 1/A \propto 1/D^2$. Incompressible flow.
Bernoulli's Equation	$P_1 + ½ \rho v_1^2 + \rho g y_1 = P_2 + ½ \rho v_2^2 + \rho g y_2$	Work-energy conservation along streamline. Static + Dynamic + Hydrostatic head.
Torricelli's Efflux Velocity	$v = \sqrt{(2gh)}$	Efflux speed from hole at depth h under free surface. Identical to free fall.
Torricelli Horizontal Range	$R = 2 \sqrt{[h (H - h)]}$	Maximum range $R_{max} = H$ at mid-depth $h = H/2$. Complementary depths match.
Pitot Airspeed	$v = \sqrt{[2 (P_{stag} - P_{stat}) / \rho]}$	Converts dynamic pressure difference into true vehicle airspeed.
Newton's Viscosity Law	$\tau = \eta (dv_x / dy)$	τ = shear stress (Pa), η = dynamic viscosity (Pa·s), dv/dy = shear strain rate.
Hagen-Poiseuille Law	$Q = \pi R^4 \Delta P / (8 \eta L)$	Laminar flow in pipe. Flow scales with R^4. Hydraulic resistance $R_{hyd} \propto 1/R^4$.
Reynolds Number	$Re = \rho v D / \eta = v D / \nu$	Inertia / Viscosity ratio. Re < 2000: Laminar; Re > 4000: Turbulent pipe flow.
Stokes' Creeping Drag	$F_d = 6 \pi \eta r v$	Drag on sphere of radius r at $Re \ll 1$. Terminal $v_t = 2r^2(\rho_s - \rho_f)g / (9\eta)$.
Young-Laplace Equation	$\Delta P = 2\gamma / R$ [droplet]	Surface tension γ (N/m). Soap bubble with two interfaces: $\Delta P = 4\gamma / R$.
Capillary Rise (Jurin)	$h = (2 \gamma \cos \theta) / (\rho g r)$	θ = contact angle. $\theta < 90° \Rightarrow$ capillary rise; $\theta > 90° \Rightarrow$ depression.

Appendix B: Standard Physical Constants & Fluid Properties Reference

Physical Property / Constant	Standard Symbol	Accepted Standard Value (SI Units)	Practical Usage Context
Standard Atmospheric Pressure	P_{atm} / P_0	**1.01325×10^5 Pa** = 101.325 kPa	Sea level reference, barometers (760 mmHg, 1.013 bar)
Standard Acceleration of Gravity	g	**$9.80665 m/s^2 \approx 9.80 m/s^2$**	Hydrostatic head and weight calculations
Density of Pure Fresh Water (4 °C)	ρ_{water}	**1,000 kg/m^3** = 1.000 g/cm^3	Universal specific gravity benchmark (SG = 1.000)
Density of Sea Water (3.5% salinity)	ρ_{sea}	**1,025 kg/m^3** = 1.025 g/cm^3	Marine navigation, iceberg submersion calculations

Physical Property / Constant	Standard Symbol	Accepted Standard Value (SI Units)	Practical Usage Context
Density of Liquid Mercury (0 °C)	ρ_{Hg}	**13,600 kg/m^3** = 13.60 g/cm^3	Manometers, barometers (760 mm column)
Density of Standard Air (20 °C, 1 atm)	ρ_{air}	**1.204 kg/m^3**	Aerodynamics, Pitot tubes, Reynolds numbers
Dynamic Viscosity of Water (20 °C)	η_{water}	**1.002 × 10^{-3} Pa · s** = 1.002 cP	Poiseuille pipe flow, hydraulic drag
Dynamic Viscosity of Air (20 °C)	η_{air}	**1.82 × 10^{-5} Pa · s** = 0.0182 cP	Aerodynamic boundary layers, Stokes' settling
Surface Tension of Water (20 °C in air)	γ_{water}	**0.0728 N/m** = 72.8 mN/m	Young-Laplace bubbles, Jurin capillary rise
Saturation Vapor Pressure of Water (20 °C)	P_{vap}	**2,338 Pa** = 2.34 kPa = 0.0231 atm	Cavitation threshold in siphons and Venturi throats